

3. First bud production

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1 Environment preparation

Run scripts with the needed packages, functions, and the raw data frame.

```
source("0a_packages.R")
source("0b_script_data.R")
source("0c_functions.R")
```

2 Data formatting

Create a new data frame adapted for survival analysis.

```
FB_df <-

  #' Load the raw data frame.
  df %>%

  #' Define the follow-up time for each individual:
  #' if the hydra produced a first bud -> time of first bud production (event),
  #' if the hydra died -> time of death (censor),
  #' if the hydra did not produce a first bud and stayed alive
  #' -> 42 days (no event, censor at the end of the experiment).
  mutate(age_followup_days = case_when(
    FB_state == 1 ~ age_FB,
    FB_state == 0 & death_state == 1 ~ age_death,
    TRUE ~ 42),
```

```

age_followup = age_followup_days / 7,
FB_event = FB_state) %>%

#' Build a competing-risk variable for the cumulative incidence functions:
#' if the hydra did not produce a first bud and stayed alive -> 0 (no event),
#' if the hydra produced a first bud -> 1 (event),
#' if the hydra died -> 2 (competing risk of death).
mutate(status_competing = factor(case_when(FB_event == 1 ~ 1,
                                           death_state == 1 ~ 2,
                                           TRUE ~ 0))) %>%

#' Build the variable tum_state_early: an individual is considered
#' tumoral only if the tumor appeared before or at the same time as the
#' supernumerary tentacle event.
mutate(age_tum_weeks = age_tum / 7,
       tum_state_early = case_when(tum_state == "Tumoral" &
                                   !is.na(age_tum_weeks) &
                                   age_tum_weeks <= age_followup ~ "Tumoral",
                                   TRUE ~ "Healthy"),
       tum_state_early = factor(tum_state_early, levels = c("Healthy", "Tumoral"))) %>%

#' Remove missing data.
filter(!is.na(age_followup)) %>%

#' Keep only the variables needed for the analysis.
select(age_followup, FB_event, status_competing, rest_condition, lineage,
       tum_state_early, parent_unique, replicate_unique)

```

Set reference levels for the categorical variables.

```

FB_df$lineage <- relevel(factor(FB_df$lineage), ref = "HO_MT")
FB_df$rest_condition <- relevel(factor(FB_df$rest_condition), ref = "Control")

```

3 Survival object

Create a survival object with follow-up time and event indicator.

```

FB_surv_object <- Surv(FB_df$age_followup, FB_df$FB_event)

```

4 Proportional hazards assumption check : Schoenfeld residuals

Fit a preliminary full Cox model (fixed effects only).

```

FB_cox <- coxph(Surv(age_followup, FB_event) ~ rest_condition * lineage *
               tum_state_early, data = FB_df)

```

Test the proportional hazards assumption.

```
print(cox.zph(FB_cox))
```

```
##                chisq df      p
## rest_condition      5.55  1  0.019
## lineage            131.26  5 < 2e-16
## tum_state_early      4.39  1  0.036
## rest_condition:lineage  55.59  5 9.9e-11
## rest_condition:tum_state_early  6.19  1  0.013
## lineage:tum_state_early  5.44  2  0.066
## rest_condition:lineage:tum_state_early  2.50  1  0.114
## GLOBAL              176.70 16 < 2e-16
```

The proportional hazards assumption is violated for at least one term, an accelerated failure time mixed-effects model will be used.

5 AIC comparison of different distributions

Compare intercept-only accelerated failure time models fitted with different parametric distributions to identify the best-fitting one.

```
FB_dist_weib <- SurvregME(Surv(age_followup, FB_event) ~ 1, data = FB_df,
                          dist = "weibull")

FB_dist_lnorm <- SurvregME(Surv(age_followup, FB_event) ~ 1, data = FB_df,
                           dist = "lognormal")

FB_dist_llog <- SurvregME(Surv(age_followup, FB_event) ~ 1, data = FB_df,
                           dist = "loglogistic")

FB_dist_exp <- SurvregME(Surv(age_followup, FB_event) ~ 1, data = FB_df,
                          dist = "exponential")

AIC(FB_dist_weib, FB_dist_lnorm, FB_dist_llog, FB_dist_exp) %>% arrange(AIC)
```

```
##          df      AIC
## FB_dist_lnorm  2 2521.574
## FB_dist_llog   2 2530.357
## FB_dist_weib   2 2634.431
## FB_dist_exp    1 2648.340
```

The model with a log-normal distribution has the lowest AIC, we'll use it for the models.

6 Model selection

6.1 Random effects selection

With the full fixed-effect structure held constant, compare different random-effect structures to find the optimal one.

```

FB_R_0 <- SurvregME(Surv(age_followup, FB_event) ~
  rest_condition * lineage * tum_state_early,
  data = FB_df, dist = "lognormal")

FB_R_1 <- SurvregME(Surv(age_followup, FB_event) ~
  rest_condition * lineage * tum_state_early +
  (1 | parent_unique),
  data = FB_df, dist = "lognormal")

FB_R_2 <- SurvregME(Surv(age_followup, FB_event) ~
  rest_condition * lineage * tum_state_early +
  (1 | replicate_unique),
  data = FB_df, dist = "lognormal")

FB_R_3 <- SurvregME(Surv(age_followup, FB_event) ~
  rest_condition * lineage * tum_state_early +
  (1 | parent_unique) + (1 | replicate_unique),
  data = FB_df, dist = "lognormal")

AIC(FB_R_0, FB_R_1, FB_R_2, FB_R_3) %>% arrange(AIC)

```

```

##      df      AIC
## FB_R_2 26 2041.720
## FB_R_3 27 2043.720
## FB_R_0 25 2197.253
## FB_R_1 26 2199.253

```

The model with the replicate id as a random effect (FB_R_2) has the lowest AIC, this random effect structure is therefore selected.

6.2 Fixed effects selection

With the selected random-effect structure fixed, compare different fixed-effect structures to find the optimal one.

```

FB_F_0 <- SurvregME(Surv(age_followup, FB_event) ~ 1 + (1 | replicate_unique), data =
  FB_df, dist = "lognormal")
FB_F_1 <- SurvregME(Surv(age_followup, FB_event) ~ rest_condition + (1 |
  replicate_unique), data = FB_df, dist = "lognormal")
FB_F_2 <- SurvregME(Surv(age_followup, FB_event) ~ lineage + (1 | replicate_unique), data
  = FB_df, dist = "lognormal")
FB_F_3 <- SurvregME(Surv(age_followup, FB_event) ~ tum_state_early + (1 |
  replicate_unique), data = FB_df, dist = "lognormal")
FB_F_4 <- SurvregME(Surv(age_followup, FB_event) ~ rest_condition + lineage + (1 |
  replicate_unique), data = FB_df, dist = "lognormal")
FB_F_5 <- SurvregME(Surv(age_followup, FB_event) ~ rest_condition + tum_state_early + (1
  | replicate_unique), data = FB_df, dist = "lognormal")
FB_F_6 <- SurvregME(Surv(age_followup, FB_event) ~ lineage + tum_state_early + (1 |
  replicate_unique), data = FB_df, dist = "lognormal")
FB_F_7 <- SurvregME(Surv(age_followup, FB_event) ~ rest_condition * lineage + (1 |
  replicate_unique), data = FB_df, dist = "lognormal")
FB_F_8 <- SurvregME(Surv(age_followup, FB_event) ~ rest_condition * tum_state_early + (1
  | replicate_unique), data = FB_df, dist = "lognormal")

```

```

FB_F_9 <- SurvregME(Surv(age_followup, FB_event) ~ lineage * tum_state_early + (1 |
replicate_unique), data = FB_df, dist = "lognormal")
FB_F_10 <- SurvregME(Surv(age_followup, FB_event) ~ rest_condition + lineage +
tum_state_early + (1 | replicate_unique), data = FB_df, dist = "lognormal")
FB_F_11 <- SurvregME(Surv(age_followup, FB_event) ~ rest_condition * lineage +
tum_state_early + (1 | replicate_unique), data = FB_df, dist = "lognormal")
FB_F_12 <- SurvregME(Surv(age_followup, FB_event) ~ rest_condition * tum_state_early +
lineage + (1 | replicate_unique), data = FB_df, dist = "lognormal")
FB_F_13 <- SurvregME(Surv(age_followup, FB_event) ~ lineage * tum_state_early +
rest_condition + (1 | replicate_unique), data = FB_df, dist = "lognormal")
FB_F_14 <- SurvregME(Surv(age_followup, FB_event) ~ rest_condition * lineage *
tum_state_early + (1 | replicate_unique), data = FB_df, dist = "lognormal")

AIC(FB_F_0, FB_F_1, FB_F_2, FB_F_3, FB_F_4, FB_F_5, FB_F_6, FB_F_7,
    FB_F_8, FB_F_9, FB_F_10, FB_F_11, FB_F_12, FB_F_13, FB_F_14) %>% arrange(AIC)

```

```

##      df      AIC
## FB_F_11 15 2022.674
## FB_F_12 11 2039.850
## FB_F_10 10 2040.381
## FB_F_14 26 2041.720
## FB_F_6   9 2044.891
## FB_F_13 15 2047.613
## FB_F_9   14 2051.542
## FB_F_7   14 2114.692
## FB_F_4    9 2123.404
## FB_F_2    8 2132.611
## FB_F_8    6 2207.268
## FB_F_5    5 2209.009
## FB_F_3    4 2214.100
## FB_F_1    4 2266.732
## FB_F_0    3 2275.753

```

The model with the interaction between the rest condition and the lineage, and with the additive effect of tumoral state has the lowest AIC and is therefore selected (FB_F_11).

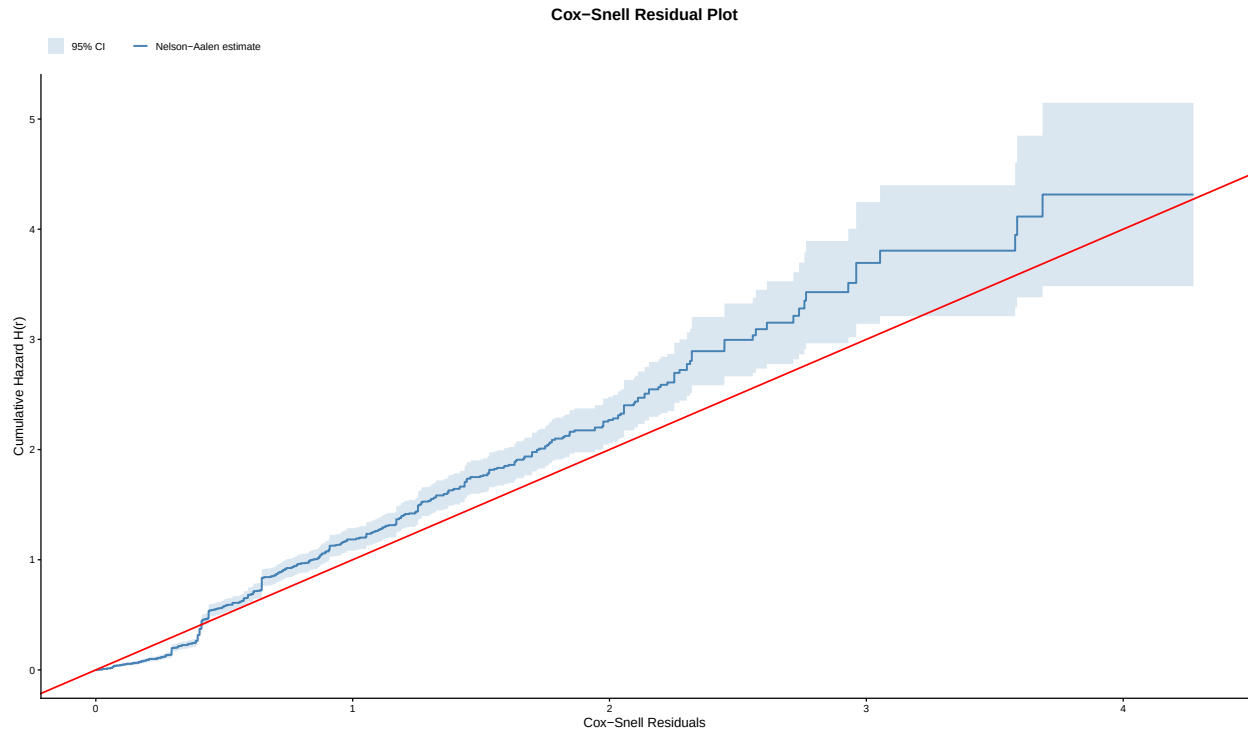
6.3 Goodness-of-fit evaluation

Assess goodness-of-fit of the selected model using Cox-Snell residuals.

```

plot_cox_snell_survregme(model = FB_F_11, data = FB_df, time_col = "age_followup",
event_col = "FB_event")

```



The curve follows a 45 degree exponential, we validate the fit of the selected model.

6.4 Selected model summary

#' Releveling of "lineage" to obtain all pairwise comparisons from the same model.

```
FB_df$lineage <- relevel(FB_df$lineage, ref = "HO_MT")
summary(SurvregME(Surv(age_followup, FB_event) ~
  rest_condition * lineage + tum_state_early + (1 | replicate_unique),
  data = FB_df, dist = "lognormal"))
```

```
##
## Mixed-Effects Log-Normal Linear Regression Model
##
## Formula:
## Surv(age_followup, FB_event) ~ rest_condition * lineage + tum_state_early +
##   (1 | replicate_unique)
##
## Fitted to dataset FB_df
##
## Fixed effects parameters:
## =====
##
##
```

	Estimate	Std. Error	z value
rest_conditionMechanical disruption	0.048981	0.170051	0.2880
lineageHC_MT	2.211147	0.298500	7.4075
lineageHO_SPC	0.185987	0.257831	0.7214
lineageHO_SPT	1.474533	0.266992	5.5228
lineageHO_VLN	2.069535	0.252228	8.2050

```

## lineageHV_GAL 1.971254 0.267067 7.3811
## tum_state_earlyTumoral 2.234789 0.234979 9.5106
## rest_conditionMechanical disruption:lineageHC_MT 0.578781 0.269384 2.1485
## rest_conditionMechanical disruption:lineageHO_SPC -0.036154 0.238204 -0.1518
## rest_conditionMechanical disruption:lineageHO_SPT -0.516544 0.242203 -2.1327
## rest_conditionMechanical disruption:lineageHO_VLN 0.547600 0.225972 2.4233
## rest_conditionMechanical disruption:lineageHV_GAL 0.212865 0.239548 0.8886
## Pr(>|z|)
## rest_conditionMechanical disruption 0.77332
## lineageHC_MT 1.287e-13 ***
## lineageHO_SPC 0.47069
## lineageHO_SPT 3.337e-08 ***
## lineageHO_VLN 2.306e-16 ***
## lineageHV_GAL 1.570e-13 ***
## tum_state_earlyTumoral < 2.2e-16 ***
## rest_conditionMechanical disruption:lineageHC_MT 0.03167 *
## rest_conditionMechanical disruption:lineageHO_SPC 0.87936
## rest_conditionMechanical disruption:lineageHO_SPT 0.03295 *
## rest_conditionMechanical disruption:lineageHO_VLN 0.01538 *
## rest_conditionMechanical disruption:lineageHV_GAL 0.37421
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Random effects:
## =====
##
## Grouping factor: replicate_unique (432 levels)
## Standard deviation:
## (Intercept)
## 1.1703
##
## Log-likelihood: -996.3371 (npar = 15)

```

```

FB_df$lineage <- relevel(FB_df$lineage, ref = "HO_SPC")
summary(SurvregME(Surv(age_followup, FB_event) ~
  rest_condition * lineage + tum_state_early + (1 | replicate_unique),
  data = FB_df, dist = "lognormal"))

```

```

##
## Mixed-Effects Log-Normal Linear Regression Model
##
## Formula:
## Surv(age_followup, FB_event) ~ rest_condition * lineage + tum_state_early +
## (1 | replicate_unique)
##
## Fitted to dataset FB_df
##
## Fixed effects parameters:
## =====
##
## Estimate Std. Error z value
## rest_conditionMechanical disruption 0.012831 0.167869 0.0764

```

```
## lineageHO_MT -0.185979 0.257831 -0.7213
## lineageHC_MT 2.025164 0.296385 6.8329
## lineageHO_SPT 1.288545 0.265838 4.8471
## lineageHO_VLN 1.883548 0.249845 7.5389
## lineageHV_GAL 1.785271 0.264853 6.7406
## tum_state_earlyTumoral 2.234794 0.234979 9.5106
## rest_conditionMechanical disruption:lineageHO_MT 0.036143 0.238204 0.1517
## rest_conditionMechanical disruption:lineageHC_MT 0.614928 0.267964 2.2948
## rest_conditionMechanical disruption:lineageHO_SPT -0.480394 0.240519 -1.9973
## rest_conditionMechanical disruption:lineageHO_VLN 0.583752 0.224262 2.6030
## rest_conditionMechanical disruption:lineageHV_GAL 0.249012 0.237980 1.0464
## Pr(>|z|)
## rest_conditionMechanical disruption 0.939071
## lineageHO_MT 0.470712
## lineageHC_MT 8.322e-12 ***
## lineageHO_SPT 1.253e-06 ***
## lineageHO_VLN 4.741e-14 ***
## lineageHV_GAL 1.577e-11 ***
## tum_state_earlyTumoral < 2.2e-16 ***
## rest_conditionMechanical disruption:lineageHO_MT 0.879400
## rest_conditionMechanical disruption:lineageHC_MT 0.021744 *
## rest_conditionMechanical disruption:lineageHO_SPT 0.045790 *
## rest_conditionMechanical disruption:lineageHO_VLN 0.009242 **
## rest_conditionMechanical disruption:lineageHV_GAL 0.295396
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Random effects:
## =====
##
## Grouping factor: replicate_unique (432 levels)
## Standard deviation:
## (Intercept)
## 1.1703
##
##
## Log-likelihood: -996.3371 (npar = 15)
```

```
FB_df$lineage <- relevel(FB_df$lineage, ref = "HO_SPT")
summary(SurvregME(Surv(age_followup, FB_event) ~
  rest_condition * lineage + tum_state_early + (1 | replicate_unique),
  data = FB_df, dist = "lognormal"))
```

```
##
## Mixed-Effects Log-Normal Linear Regression Model
##
## Formula:
## Surv(age_followup, FB_event) ~ rest_condition * lineage + tum_state_early +
## (1 | replicate_unique)
##
## Fitted to dataset FB_df
##
## Fixed effects parameters:
```

```

## =====
##
##
## Estimate Std. Error z value
## rest_conditionMechanical disruption -0.46757 0.17350 -2.6949
## lineageHO_SPC -1.28855 0.26584 -4.8471
## lineageHO_MT -1.47453 0.26699 -5.5228
## lineageHC_MT 0.73661 0.29238 2.5194
## lineageHO_VLN 0.59499 0.24624 2.4163
## lineageHV_GAL 0.49671 0.26189 1.8966
## tum_state_earlyTumoral 2.23480 0.23498 9.5106
## rest_conditionMechanical disruption:lineageHO_SPC 0.48040 0.24052 1.9974
## rest_conditionMechanical disruption:lineageHO_MT 0.51654 0.24220 2.1327
## rest_conditionMechanical disruption:lineageHC_MT 1.09533 0.27264 4.0176
## rest_conditionMechanical disruption:lineageHO_VLN 1.06416 0.22974 4.6319
## rest_conditionMechanical disruption:lineageHV_GAL 0.72942 0.24241 3.0090
## Pr(>|z|)
## rest_conditionMechanical disruption 0.007041 **
## lineageHO_SPC 1.253e-06 ***
## lineageHO_MT 3.337e-08 ***
## lineageHC_MT 0.011756 *
## lineageHO_VLN 0.015678 *
## lineageHV_GAL 0.057877 .
## tum_state_earlyTumoral < 2.2e-16 ***
## rest_conditionMechanical disruption:lineageHO_SPC 0.045787 *
## rest_conditionMechanical disruption:lineageHO_MT 0.032950 *
## rest_conditionMechanical disruption:lineageHC_MT 5.880e-05 ***
## rest_conditionMechanical disruption:lineageHO_VLN 3.623e-06 ***
## rest_conditionMechanical disruption:lineageHV_GAL 0.002621 **
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Random effects:
## =====
##
## Grouping factor: replicate_unique (432 levels)
## Standard deviation:
## (Intercept)
## 1.1703
##
##
## Log-likelihood: -996.3371 (npar = 15)

FB_df$lineage <- relevel(FB_df$lineage, ref = "HO_VLN")
summary(SurvregME(Surv(age_followup, FB_event) ~
  rest_condition * lineage + tum_state_early + (1 | replicate_unique),
  data = FB_df, dist = "lognormal"))

##
## Mixed-Effects Log-Normal Linear Regression Model
##
## Formula:
## Surv(age_followup, FB_event) ~ rest_condition * lineage + tum_state_early +
## (1 | replicate_unique)

```

```

##
## Fitted to dataset FB_df
##
## Fixed effects parameters:
## =====
##
##                                     Estimate Std. Error z value
## rest_conditionMechanical disruption      0.596595    0.149017  4.0035
## lineageHO_SPT                          -0.594993    0.246240 -2.4163
## lineageHO_SPC                          -1.883540    0.249845 -7.5388
## lineageHO_MT                           -2.069524    0.252228 -8.2050
## lineageHC_MT                            0.141629    0.272716  0.5193
## lineageHV_GAL                          -0.098264    0.240411 -0.4087
## tum_state_earlyTumoral                  2.234804    0.234979  9.5107
## rest_conditionMechanical disruption:lineageHO_SPT -1.064162    0.229745 -4.6319
## rest_conditionMechanical disruption:lineageHO_SPC -0.583767    0.224262 -2.6031
## rest_conditionMechanical disruption:lineageHO_MT -0.547622    0.225972 -2.4234
## rest_conditionMechanical disruption:lineageHC_MT  0.031167    0.255372  0.1220
## rest_conditionMechanical disruption:lineageHV_GAL -0.334754    0.224607 -1.4904
##                                     Pr(>|z|)
## rest_conditionMechanical disruption      6.240e-05 ***
## lineageHO_SPT                          0.01568 *
## lineageHO_SPC                          4.742e-14 ***
## lineageHO_MT                           2.307e-16 ***
## lineageHC_MT                            0.60353
## lineageHV_GAL                           0.68273
## tum_state_earlyTumoral                  < 2.2e-16 ***
## rest_conditionMechanical disruption:lineageHO_SPT 3.623e-06 ***
## rest_conditionMechanical disruption:lineageHO_SPC 0.00924 **
## rest_conditionMechanical disruption:lineageHO_MT 0.01538 *
## rest_conditionMechanical disruption:lineageHC_MT 0.90286
## rest_conditionMechanical disruption:lineageHV_GAL 0.13612
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Random effects:
## =====
##
## Grouping factor: replicate_unique (432 levels)
## Standard deviation:
## (Intercept)
##      1.1703
##
##
## Log-likelihood: -996.3371 (npar = 15)

FB_df$lineage <- relevel(FB_df$lineage, ref = "HC_MT")
summary(SurvregME(Surv(age_followup, FB_event) ~
  rest_condition * lineage + tum_state_early + (1 | replicate_unique),
  data = FB_df, dist = "lognormal"))

##
## Mixed-Effects Log-Normal Linear Regression Model

```

```
##
## Formula:
## Surv(age_followup, FB_event) ~ rest_condition * lineage + tum_state_early +
## (1 | replicate_unique)
##
## Fitted to dataset FB_df
##
## Fixed effects parameters:
## =====
##
##                                     Estimate Std. Error z value
## rest_conditionMechanical disruption      0.627755    0.209080  3.0025
## lineageHO_VLN                        -0.141603    0.272716 -0.5192
## lineageHO_SPT                        -0.736598    0.292376 -2.5194
## lineageHO_SPC                        -2.025149    0.296384 -6.8329
## lineageHO_MT                         -2.211131    0.298500 -7.4075
## lineageHV_GAL                       -0.239892    0.287121 -0.8355
## tum_state_earlyTumoral                2.234790    0.234979  9.5106
## rest_conditionMechanical disruption:lineageHO_VLN -0.031172    0.255372 -0.1221
## rest_conditionMechanical disruption:lineageHO_SPT -1.095320    0.272637 -4.0175
## rest_conditionMechanical disruption:lineageHO_SPC -0.614927    0.267964 -2.2948
## rest_conditionMechanical disruption:lineageHO_MT  -0.578778    0.269384 -2.1485
## rest_conditionMechanical disruption:lineageHV_GAL -0.365902    0.268223 -1.3642
##                                     Pr(>|z|)
## rest_conditionMechanical disruption      0.002678 **
## lineageHO_VLN                          0.603600
## lineageHO_SPT                          0.011757 *
## lineageHO_SPC                         8.324e-12 ***
## lineageHO_MT                         1.287e-13 ***
## lineageHV_GAL                         0.403433
## tum_state_earlyTumoral                  < 2.2e-16 ***
## rest_conditionMechanical disruption:lineageHO_VLN  0.902847
## rest_conditionMechanical disruption:lineageHO_SPT 5.882e-05 ***
## rest_conditionMechanical disruption:lineageHO_SPC  0.021744 *
## rest_conditionMechanical disruption:lineageHO_MT   0.031672 *
## rest_conditionMechanical disruption:lineageHV_GAL  0.172514
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Random effects:
## =====
##
## Grouping factor: replicate_unique (432 levels)
## Standard deviation:
## (Intercept)
##      1.1703
##
##
## Log-likelihood: -996.3371 (npar = 15)

FB_df$lineage <- relevel(FB_df$lineage, ref = "HV_GAL")
summary(SurvregME(Surv(age_followup, FB_event) ~
  rest_condition * lineage + tum_state_early + (1 | replicate_unique),
  data = FB_df, dist = "lognormal"))
```

```

##
## Mixed-Effects Log-Normal Linear Regression Model
##
## Formula:
## Surv(age_followup, FB_event) ~ rest_condition * lineage + tum_state_early +
##      (1 | replicate_unique)
##
## Fitted to dataset FB_df
##
## Fixed effects parameters:
## =====
##
##                                     Estimate Std. Error z value
## rest_conditionMechanical disruption    0.261860    0.168784  1.5515
## lineageHC_MT                          0.239910    0.287122  0.8356
## lineageHO_VLN                         0.098294    0.240411  0.4089
## lineageHO_SPT                       -0.496709    0.261894 -1.8966
## lineageHO_SPC                       -1.785257    0.264853 -6.7406
## lineageHO_MT                        -1.971238    0.267067 -7.3811
## tum_state_earlyTumoral                2.234796    0.234979  9.5106
## rest_conditionMechanical disruption:lineageHC_MT  0.365898    0.268223  1.3642
## rest_conditionMechanical disruption:lineageHO_VLN 0.334720    0.224607  1.4902
## rest_conditionMechanical disruption:lineageHO_SPT -0.729426    0.242415 -3.0090
## rest_conditionMechanical disruption:lineageHO_SPC -0.249033    0.237980 -1.0464
## rest_conditionMechanical disruption:lineageHO_MT -0.212888    0.239548 -0.8887
##                                     Pr(>|z|)
## rest_conditionMechanical disruption    0.120793
## lineageHC_MT                          0.403398
## lineageHO_VLN                         0.682642
## lineageHO_SPT                         0.057880 .
## lineageHO_SPC                        1.578e-11 ***
## lineageHO_MT                        1.570e-13 ***
## tum_state_earlyTumoral                < 2.2e-16 ***
## rest_conditionMechanical disruption:lineageHC_MT  0.172519
## rest_conditionMechanical disruption:lineageHO_VLN 0.136159
## rest_conditionMechanical disruption:lineageHO_SPT 0.002621 **
## rest_conditionMechanical disruption:lineageHO_SPC 0.295356
## rest_conditionMechanical disruption:lineageHO_MT  0.374160
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Random effects:
## =====
##
## Grouping factor: replicate_unique (432 levels)
## Standard deviation:
## (Intercept)
##      1.1703
##
##
## Log-likelihood: -996.3371 (npar = 15)

```

7 Median times

7.1 By rest condition

Estimate cumulative incidence functions for the first bud event, accounting for the competing risk of death, stratified by rest condition.

```
FB_cuminc_rest <- cuminc(Surv(age_followup, status_competing) ~ rest_condition, data = FB_df)
```

For each stratum, extract the median time to event and its 95CI.

```
FB_medians_rest <- tidy(FB_cuminc_rest) %>%
  dplyr::filter(outcome == "1") %>%
  dplyr::group_by(strata) %>%
  dplyr::arrange(time) %>%
  dplyr::summarise(
    median_time = { idx <- which(estimate >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    ci_low = { idx <- which(conf.high >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    ci_high = { idx <- which(conf.low >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    .groups = "drop")

print(FB_medians_rest)
```

```
## # A tibble: 2 x 4
##   strata                median_time ci_low ci_high
##   <fct>                <dbl>   <dbl>   <dbl>
## 1 Control                1.14     1.14     1.14
## 2 Mechanical disruption    1.14     1.14     1.29
```

7.2 By lineage

Same method as above, but stratified by lineage.

```
FB_cuminc_lineage <- cuminc(Surv(age_followup, status_competing) ~ lineage, data = FB_df)

FB_medians_lineage <- tidy(FB_cuminc_lineage) %>%
  dplyr::filter(outcome == "1") %>%
  dplyr::group_by(strata) %>%
  dplyr::arrange(time) %>%
  dplyr::summarise(
    median_time = { idx <- which(estimate >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    ci_low = { idx <- which(conf.high >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    ci_high = { idx <- which(conf.low >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    .groups = "drop")
```

```
print(FB_medians_lineage)
```

```
## # A tibble: 6 x 4
##   strata median_time ci_low ci_high
##   <fct>      <dbl>  <dbl>  <dbl>
## 1 HV_GAL      1.71    1.29    2
## 2 HC_MT        3      2.43    3
## 3 HO_VLN       1      1      2.43
## 4 HO_SPT      1.29    1.14    1.29
## 5 HO_SPC      0.286  0.286  1.14
## 6 HO_MT       0.286  0.286  1
```

7.3 By rest condition and lineage

Same method as above, but stratified by rest condition and lineage.

```
FB_cuminc_interaction <- cuminc(Surv(age_followup, status_competing) ~ rest_condition +
lineage, data = FB_df)
```

```
FB_medians_interaction <- tidy(FB_cuminc_interaction) %>%
  dplyr::filter(outcome == "1") %>%
  dplyr::group_by(strata) %>%
  dplyr::arrange(time) %>%
  dplyr::summarise(
    median_time = { idx <- which(estimate >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    ci_low = { idx <- which(conf.high >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    ci_high = { idx <- which(conf.low >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    .groups = "drop")

print(FB_medians_interaction)
```

```
## # A tibble: 12 x 4
##   strata                median_time ci_low ci_high
##   <fct>                <dbl>  <dbl>  <dbl>
## 1 Control, HV_GAL      1.71    1.71    2
## 2 Mechanical disruption, HV_GAL 1.71    1      2.43
## 3 Control, HC_MT       1.43    1.14    3
## 4 Mechanical disruption, HC_MT   3      3      3
## 5 Control, HO_VLN       1      1      1.14
## 6 Mechanical disruption, HO_VLN 1.14    1      3.14
## 7 Control, HO_SPT      1.29    1.14    1.57
## 8 Mechanical disruption, HO_SPT 1.14    1.14    1.29
## 9 Control, HO_SPC      0.286  0.286  1.14
## 10 Mechanical disruption, HO_SPC 0.286  0.286  1.14
## 11 Control, HO_MT       0.286  0.286  1
## 12 Mechanical disruption, HO_MT 0.286  0.286  1
```

7.4 By tumoral state

Same method as above, but stratified by tumoral state.

```
FB_cuminc_tum <- cuminc(Surv(age_followup, status_competing) ~ tum_state_early, data =
FB_df)

FB_medians_tum <- tidy(FB_cuminc_tum) %>%
  dplyr::filter(outcome == "1") %>%
  dplyr::group_by(strata) %>%
  dplyr::arrange(time) %>%
  dplyr::summarise(
    median_time = { idx <- which(estimate >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    ci_low = { idx <- which(conf.high >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    ci_high = { idx <- which(conf.low >= 0.5)[1]
    if (is.na(idx)) NA_real_ else time[idx] },
    .groups = "drop")

print(FB_medians_tum)
```

```
## # A tibble: 2 x 4
##   strata median_time ci_low ci_high
##   <fct>      <dbl> <dbl>   <dbl>
## 1 Healthy      1.14   1.14    1.14
## 2 Tumoral       3      2.29    NA
```

8 Cumulative incidences

8.1 By week and rest condition

Convert the continuous cumulative incidence function into a weekly summary table: for each stratum of rest condition and each week, show the estimate with its 95CI.

```
FB_incidence_rest <- tidy(FB_cuminc_rest) %>%
  dplyr::filter(outcome == "1") %>%
  dplyr::mutate(week = ceiling(time)) %>%
  dplyr::group_by(strata, week) %>%
  dplyr::filter(time == max(time)) %>%
  dplyr::slice(1) %>%
  dplyr::ungroup() %>%
  dplyr::select(strata, week, estimate_cuminc = estimate, conf.low, conf.high) %>%
  dplyr::arrange(strata, week)

print(FB_incidence_rest)
```

```
## # A tibble: 14 x 5
##   strata      week estimate_cuminc conf.low conf.high
##   <fct>    <dbl>         <dbl>   <dbl>   <dbl>
## 1 Control      0           0      NA      NA
```

##	2	Control	1	0.433	0.386	0.479
##	3	Control	2	0.736	0.692	0.775
##	4	Control	3	0.894	0.860	0.919
##	5	Control	4	0.907	0.876	0.931
##	6	Control	5	0.931	0.902	0.951
##	7	Control	6	0.947	0.921	0.964
##	8	Mechanical disruption	0	0	NA	NA
##	9	Mechanical disruption	1	0.421	0.374	0.467
##	10	Mechanical disruption	2	0.650	0.603	0.693
##	11	Mechanical disruption	3	0.789	0.748	0.825
##	12	Mechanical disruption	4	0.822	0.782	0.855
##	13	Mechanical disruption	5	0.880	0.845	0.907
##	14	Mechanical disruption	6	0.882	0.848	0.909

8.2 By week and lineage

Same method as above, but stratified by lineage.

```
FB_incidence_lineage <- tidy(FB_cuminc_lineage) %>%
  dplyr::filter(outcome == "1") %>%
  dplyr::mutate(week = ceiling(time)) %>%
  dplyr::group_by(strata, week) %>%
  dplyr::filter(time == max(time)) %>%
  dplyr::slice(1) %>%
  dplyr::ungroup() %>%
  dplyr::select(strata, week, estimate_cuminc = estimate, conf.low, conf.high) %>%
  dplyr::arrange(strata, week)

print(FB_incidence_lineage)
```

```
## # A tibble: 42 x 5
##   strata week estimate_cuminc conf.low conf.high
##   <fct> <dbl>         <dbl>    <dbl>    <dbl>
## 1 HV_GAL  0           0         NA         NA
## 2 HV_GAL  1         0.319    0.245    0.396
## 3 HV_GAL  2         0.597    0.512    0.672
## 4 HV_GAL  3         0.868    0.800    0.914
## 5 HV_GAL  4         0.875    0.808    0.920
## 6 HV_GAL  5         0.931    0.874    0.962
## 7 HV_GAL  6         0.938    0.882    0.967
## 8 HC_MT   0           0         NA         NA
## 9 HC_MT   1        0.0729    0.0320    0.137
## 10 HC_MT  2         0.375    0.279    0.471
## 11 HC_MT  3         0.844    0.754    0.903
## 12 HC_MT  4         0.917    0.838    0.958
## 13 HC_MT  5         0.938    0.862    0.972
## 14 HC_MT  6         0.938    0.862    0.972
## 15 HO_VLN  0           0         NA         NA
## 16 HO_VLN  1         0.536    0.463    0.604
## 17 HO_VLN  2         0.573    0.500    0.640
## 18 HO_VLN  3         0.672    0.600    0.734
## 19 HO_VLN  4         0.703    0.633    0.763
## 20 HO_VLN  5         0.813    0.749    0.861
```

```
## 21 HO_VLN      6      0.849    0.789    0.893
## 22 HO_SPT      0      0      NA      NA
## 23 HO_SPT      1      0.215    0.152    0.286
## 24 HO_SPT      2      0.819    0.746    0.874
## 25 HO_SPT      3      0.868    0.800    0.914
## 26 HO_SPT      4      0.882    0.816    0.925
## 27 HO_SPT      5      0.882    0.816    0.925
## 28 HO_SPT      6      0.882    0.816    0.925
## 29 HO_SPC      0      0      NA      NA
## 30 HO_SPC      1      0.549    0.463    0.626
## 31 HO_SPC      2      0.882    0.816    0.925
## 32 HO_SPC      3      0.924    0.865    0.957
## 33 HO_SPC      4      0.951    0.899    0.977
## 34 HO_SPC      5      0.979    0.931    0.994
## 35 HO_SPC      6      0.979    0.931    0.994
## 36 HO_MT       0      0      NA      NA
## 37 HO_MT       1      0.715    0.634    0.781
## 38 HO_MT       2      0.847    0.778    0.896
## 39 HO_MT       3      0.931    0.874    0.962
## 40 HO_MT       4      0.931    0.874    0.962
## 41 HO_MT       5      0.931    0.874    0.962
## 42 HO_MT       6      0.931    0.874    0.962
```

8.3 By week, rest condition, and lineage

Same method as above, but stratified by rest condition and lineage.

```
FB_incidence_interaction <- tidy(FB_cuminc_interaction) %>%
  dplyr::filter(outcome == "1") %>%
  dplyr::mutate(week = ceiling(time)) %>%
  dplyr::group_by(strata, week) %>%
  dplyr::filter(time == max(time)) %>%
  dplyr::slice(1) %>%
  dplyr::ungroup() %>%
  dplyr::select(strata, week, estimate_cuminc = estimate, conf.low, conf.high) %>%
  dplyr::arrange(strata, week)

print(FB_incidence_interaction)
```

```
## # A tibble: 84 x 5
##   strata                                week estimate_cuminc conf.low conf.high
##   <fct>                                <dbl>         <dbl>     <dbl>     <dbl>
## 1 Control, HV_GAL                      0             0      NA      NA
## 2 Control, HV_GAL                      1          0.222    0.134    0.324
## 3 Control, HV_GAL                      2          0.639    0.515    0.739
## 4 Control, HV_GAL                      3          0.986    0.898    0.998
## 5 Control, HV_GAL                      4            NA      NA      NA
## 6 Control, HV_GAL                      5            NA      NA      NA
## 7 Control, HV_GAL                      6            NA      NA      NA
## 8 Mechanical disruption, HV_GAL        0             0      NA      NA
## 9 Mechanical disruption, HV_GAL        1          0.417    0.302    0.528
## 10 Mechanical disruption, HV_GAL       2          0.556    0.432    0.662
## 11 Mechanical disruption, HV_GAL       3          0.75     0.631    0.835
```

## 12	Mechanical disruption, HV_GAL	4	0.75	0.631	0.835
## 13	Mechanical disruption, HV_GAL	5	0.861	0.755	0.924
## 14	Mechanical disruption, HV_GAL	6	0.875	0.771	0.934
## 15	Control, HC_MT	0	0	NA	NA
## 16	Control, HC_MT	1	0.0833	0.0263	0.183
## 17	Control, HC_MT	2	0.521	0.370	0.652
## 18	Control, HC_MT	3	0.917	0.788	0.969
## 19	Control, HC_MT	4	0.979	0.777	0.998
## 20	Control, HC_MT	5	0.979	0.777	0.998
## 21	Control, HC_MT	6	0.979	0.777	0.998
## 22	Mechanical disruption, HC_MT	0	0	NA	NA
## 23	Mechanical disruption, HC_MT	1	0.0625	0.0160	0.156
## 24	Mechanical disruption, HC_MT	2	0.229	0.122	0.356
## 25	Mechanical disruption, HC_MT	3	0.771	0.623	0.867
## 26	Mechanical disruption, HC_MT	4	0.854	0.713	0.929
## 27	Mechanical disruption, HC_MT	5	0.896	0.759	0.957
## 28	Mechanical disruption, HC_MT	6	0.896	0.759	0.957
## 29	Control, HO_VLN	0	0	NA	NA
## 30	Control, HO_VLN	1	0.604	0.499	0.694
## 31	Control, HO_VLN	2	0.635	0.530	0.723
## 32	Control, HO_VLN	3	0.75	0.650	0.825
## 33	Control, HO_VLN	4	0.771	0.672	0.843
## 34	Control, HO_VLN	5	0.865	0.776	0.920
## 35	Control, HO_VLN	6	0.938	0.863	0.972
## 36	Mechanical disruption, HO_VLN	0	0	NA	NA
## 37	Mechanical disruption, HO_VLN	1	0.469	0.366	0.565
## 38	Mechanical disruption, HO_VLN	2	0.510	0.406	0.606
## 39	Mechanical disruption, HO_VLN	3	0.594	0.488	0.685
## 40	Mechanical disruption, HO_VLN	4	0.635	0.530	0.723
## 41	Mechanical disruption, HO_VLN	5	0.760	0.661	0.835
## 42	Mechanical disruption, HO_VLN	6	0.760	0.661	0.835
## 43	Control, HO_SPT	0	0	NA	NA
## 44	Control, HO_SPT	1	0.194	0.112	0.293
## 45	Control, HO_SPT	2	0.819	0.707	0.892
## 46	Control, HO_SPT	3	0.847	0.738	0.914
## 47	Control, HO_SPT	4	0.847	0.738	0.914
## 48	Control, HO_SPT	5	0.847	0.738	0.914
## 49	Control, HO_SPT	6	0.847	0.738	0.914
## 50	Mechanical disruption, HO_SPT	0	0	NA	NA
## 51	Mechanical disruption, HO_SPT	1	0.236	0.145	0.340
## 52	Mechanical disruption, HO_SPT	2	0.819	0.707	0.892
## 53	Mechanical disruption, HO_SPT	3	0.889	0.786	0.944
## 54	Mechanical disruption, HO_SPT	4	0.917	0.818	0.963
## 55	Mechanical disruption, HO_SPT	5	0.917	0.818	0.963
## 56	Mechanical disruption, HO_SPT	6	0.917	0.818	0.963
## 57	Control, HO_SPC	0	0	NA	NA
## 58	Control, HO_SPC	1	0.542	0.419	0.649
## 59	Control, HO_SPC	2	0.889	0.787	0.944
## 60	Control, HO_SPC	3	0.972	0.870	0.994
## 61	Control, HO_SPC	4	0.972	0.870	0.994
## 62	Control, HO_SPC	5	NA	NA	NA
## 63	Control, HO_SPC	6	NA	NA	NA
## 64	Mechanical disruption, HO_SPC	0	0	NA	NA
## 65	Mechanical disruption, HO_SPC	1	0.556	0.433	0.662

## 66	Mechanical disruption, HO_SPC	2	0.875	0.771	0.934
## 67	Mechanical disruption, HO_SPC	3	0.875	0.771	0.934
## 68	Mechanical disruption, HO_SPC	4	0.931	0.836	0.972
## 69	Mechanical disruption, HO_SPC	5	0.972	0.874	0.994
## 70	Mechanical disruption, HO_SPC	6	0.972	0.874	0.994
## 71	Control, HO_MT	0	0	NA	NA
## 72	Control, HO_MT	1	0.778	0.663	0.858
## 73	Control, HO_MT	2	0.875	0.771	0.934
## 74	Control, HO_MT	3	0.944	0.852	0.980
## 75	Control, HO_MT	4	0.944	0.852	0.980
## 76	Control, HO_MT	5	0.944	0.852	0.980
## 77	Control, HO_MT	6	0.944	0.852	0.980
## 78	Mechanical disruption, HO_MT	0	0	NA	NA
## 79	Mechanical disruption, HO_MT	1	0.653	0.530	0.751
## 80	Mechanical disruption, HO_MT	2	0.819	0.709	0.891
## 81	Mechanical disruption, HO_MT	3	0.917	0.821	0.962
## 82	Mechanical disruption, HO_MT	4	0.917	0.821	0.962
## 83	Mechanical disruption, HO_MT	5	0.917	0.821	0.962
## 84	Mechanical disruption, HO_MT	6	0.917	0.821	0.962

8.4 By week and tumoral state

Same method as above, but stratified by tumoral state.

```
FB_incidence_tum <- tidy(FB_cuminc_tum) %>%
  dplyr::filter(outcome == "1") %>%
  dplyr::mutate(week = ceiling(time)) %>%
  dplyr::group_by(strata, week) %>%
  dplyr::filter(time == max(time)) %>%
  dplyr::slice(1) %>%
  dplyr::ungroup() %>%
  dplyr::select(strata, week, estimate_cuminc = estimate, conf.low, conf.high) %>%
  dplyr::arrange(strata, week)

print(FB_incidence_tum)
```

```
## # A tibble: 14 x 5
##   strata week estimate_cuminc conf.low conf.high
##   <fct>  <dbl>         <dbl>    <dbl>    <dbl>
## 1 Healthy  0           0      NA      NA
## 2 Healthy  1         0.447  0.413  0.481
## 3 Healthy  2         0.712  0.680  0.742
## 4 Healthy  3         0.858  0.832  0.880
## 5 Healthy  4         0.880  0.856  0.900
## 6 Healthy  5         0.919  0.898  0.935
## 7 Healthy  6         0.928  0.908  0.944
## 8 Tumoral  0           0      NA      NA
## 9 Tumoral  1        0.0244 0.00185 0.112
##10 Tumoral  2        0.317  0.181  0.462
##11 Tumoral  3        0.512  0.349  0.654
##12 Tumoral  4        0.561  0.394  0.699
##13 Tumoral  5        0.634  0.464  0.763
##14 Tumoral  6        0.634  0.464  0.763
```

9 Cumulative proportions

9.1 By week and rest condition

Compute raw cumulative proportions of first bud production per week, stratified by rest condition.

```
FB_proportions_rest <- FB_df %>%
  dplyr::mutate(week_FB_event = ceiling(age_followup)) %>%
  dplyr::group_by(rest_condition) %>%
  dplyr::summarise(total_N = n(),
    n1 = sum(week_FB_event <= 1 & FB_event == 1),
    n2 = sum(week_FB_event <= 2 & FB_event == 1),
    n3 = sum(week_FB_event <= 3 & FB_event == 1),
    n4 = sum(week_FB_event <= 4 & FB_event == 1),
    n5 = sum(week_FB_event <= 5 & FB_event == 1),
    n6 = sum(week_FB_event <= 6 & FB_event == 1),
    p1 = n1 / total_N,
    p2 = n2 / total_N,
    p3 = n3 / total_N,
    p4 = n4 / total_N,
    p5 = n5 / total_N,
    p6 = n6 / total_N,
    .groups = "drop") %>%
  tidyr::pivot_longer(cols = matches("[np][1-6]$"),
    names_to = c(".value", "week"),
    names_pattern = "(n|p)([1-6])") %>%
  dplyr::mutate(week = as.numeric(week)) %>%
  dplyr::select(rest_condition, week, n, total_N, raw_proportion = p)

print(FB_proportions_rest)
```

```
## # A tibble: 12 x 5
##   rest_condition      week      n total_N raw_proportion
##   <fct>          <dbl> <int>   <int>         <dbl>
## 1 Control           1    187    432         0.433
## 2 Control           2    318    432         0.736
## 3 Control           3    386    432         0.894
## 4 Control           4    392    432         0.907
## 5 Control           5    402    432         0.931
## 6 Control           6    409    432         0.947
## 7 Mechanical disruption 1    182    432         0.421
## 8 Mechanical disruption 2    281    432         0.650
## 9 Mechanical disruption 3    341    432         0.789
## 10 Mechanical disruption 4    355    432         0.822
## 11 Mechanical disruption 5    380    432         0.880
## 12 Mechanical disruption 6    381    432         0.882
```

9.2 By week and lineage

Same method as above, but stratified by lineage.

```

FB_proportions_lineage <- FB_df %>%
  dplyr::mutate(week_FB_event = ceiling(age_followup)) %>%
  dplyr::group_by(lineage) %>%
  dplyr::summarise(total_N = n(),
    n1 = sum(week_FB_event <= 1 & FB_event == 1),
    n2 = sum(week_FB_event <= 2 & FB_event == 1),
    n3 = sum(week_FB_event <= 3 & FB_event == 1),
    n4 = sum(week_FB_event <= 4 & FB_event == 1),
    n5 = sum(week_FB_event <= 5 & FB_event == 1),
    n6 = sum(week_FB_event <= 6 & FB_event == 1),
    p1 = n1 / total_N,
    p2 = n2 / total_N,
    p3 = n3 / total_N,
    p4 = n4 / total_N,
    p5 = n5 / total_N,
    p6 = n6 / total_N,
    .groups = "drop") %>%
  tidyr::pivot_longer(cols = matches("^np[1-6]$"),
    names_to = c(".value", "week"),
    names_pattern = "(n|p)([1-6])") %>%
  dplyr::mutate(week = as.numeric(week)) %>%
  dplyr::select(lineage, week, n, total_N, raw_proportion = p)

print(FB_proportions_lineage)

```

```

## # A tibble: 36 x 5
##   lineage week    n total_N raw_proportion
##   <fct>   <dbl> <int>   <int>         <dbl>
## 1 HV_GAL     1    46    144         0.319
## 2 HV_GAL     2    86    144         0.597
## 3 HV_GAL     3   125    144         0.868
## 4 HV_GAL     4   126    144         0.875
## 5 HV_GAL     5   134    144         0.931
## 6 HV_GAL     6   135    144         0.938
## 7 HC_MT      1     7     96         0.0729
## 8 HC_MT      2    36     96         0.375
## 9 HC_MT      3    81     96         0.844
## 10 HC_MT     4    88     96         0.917
## 11 HC_MT     5    90     96         0.938
## 12 HC_MT     6    90     96         0.938
## 13 HO_VLN     1   103    192         0.536
## 14 HO_VLN     2   110    192         0.573
## 15 HO_VLN     3   129    192         0.672
## 16 HO_VLN     4   135    192         0.703
## 17 HO_VLN     5   156    192         0.812
## 18 HO_VLN     6   163    192         0.849
## 19 HO_SPT     1    31    144         0.215
## 20 HO_SPT     2   118    144         0.819
## 21 HO_SPT     3   125    144         0.868
## 22 HO_SPT     4   127    144         0.882
## 23 HO_SPT     5   127    144         0.882
## 24 HO_SPT     6   127    144         0.882
## 25 HO_SPC     1    79    144         0.549
## 26 HO_SPC     2   127    144         0.882

```

## 27	HO_SPC	3	133	144	0.924
## 28	HO_SPC	4	137	144	0.951
## 29	HO_SPC	5	141	144	0.979
## 30	HO_SPC	6	141	144	0.979
## 31	HO_MT	1	103	144	0.715
## 32	HO_MT	2	122	144	0.847
## 33	HO_MT	3	134	144	0.931
## 34	HO_MT	4	134	144	0.931
## 35	HO_MT	5	134	144	0.931
## 36	HO_MT	6	134	144	0.931

9.3 By week, rest condition, and lineage

Same method as above, but stratified by rest condition and lineage.

```
FB_proportions_interaction <- FB_df %>%
  dplyr::mutate(week_FB_event = ceiling(age_followup)) %>%
  dplyr::group_by(rest_condition, lineage) %>%
  dplyr::summarise(total_N = n(),
    n1 = sum(week_FB_event <= 1 & FB_event == 1),
    n2 = sum(week_FB_event <= 2 & FB_event == 1),
    n3 = sum(week_FB_event <= 3 & FB_event == 1),
    n4 = sum(week_FB_event <= 4 & FB_event == 1),
    n5 = sum(week_FB_event <= 5 & FB_event == 1),
    n6 = sum(week_FB_event <= 6 & FB_event == 1),
    p1 = n1 / total_N,
    p2 = n2 / total_N,
    p3 = n3 / total_N,
    p4 = n4 / total_N,
    p5 = n5 / total_N,
    p6 = n6 / total_N,
    .groups = "drop") %>%
  tidyr::pivot_longer(cols = matches("^np[1-6]$",
    names_to = c(".value", "week"),
    names_pattern = "(n|p)([1-6])") %>%
  dplyr::mutate(week = as.numeric(week)) %>%
  dplyr::select(rest_condition, lineage, week, n, total_N, raw_proportion = p)

print(FB_proportions_interaction)
```

```
## # A tibble: 72 x 6
##   rest_condition lineage week  n total_N raw_proportion
##   <fct>          <fct>  <dbl> <int>  <int>      <dbl>
## 1 Control      HV_GAL    1    16    72      0.222
## 2 Control      HV_GAL    2    46    72      0.639
## 3 Control      HV_GAL    3    71    72      0.986
## 4 Control      HV_GAL    4    72    72      1
## 5 Control      HV_GAL    5    72    72      1
## 6 Control      HV_GAL    6    72    72      1
## 7 Control      HC_MT     1     4    48     0.0833
## 8 Control      HC_MT     2    25    48     0.521
## 9 Control      HC_MT     3    44    48     0.917
## 10 Control     HC_MT     4    47    48     0.979
```

## 11 Control	HC_MT	5	47	48	0.979
## 12 Control	HC_MT	6	47	48	0.979
## 13 Control	HO_VLN	1	58	96	0.604
## 14 Control	HO_VLN	2	61	96	0.635
## 15 Control	HO_VLN	3	72	96	0.75
## 16 Control	HO_VLN	4	74	96	0.771
## 17 Control	HO_VLN	5	83	96	0.865
## 18 Control	HO_VLN	6	90	96	0.938
## 19 Control	HO_SPT	1	14	72	0.194
## 20 Control	HO_SPT	2	59	72	0.819
## 21 Control	HO_SPT	3	61	72	0.847
## 22 Control	HO_SPT	4	61	72	0.847
## 23 Control	HO_SPT	5	61	72	0.847
## 24 Control	HO_SPT	6	61	72	0.847
## 25 Control	HO_SPC	1	39	72	0.542
## 26 Control	HO_SPC	2	64	72	0.889
## 27 Control	HO_SPC	3	70	72	0.972
## 28 Control	HO_SPC	4	70	72	0.972
## 29 Control	HO_SPC	5	71	72	0.986
## 30 Control	HO_SPC	6	71	72	0.986
## 31 Control	HO_MT	1	56	72	0.778
## 32 Control	HO_MT	2	63	72	0.875
## 33 Control	HO_MT	3	68	72	0.944
## 34 Control	HO_MT	4	68	72	0.944
## 35 Control	HO_MT	5	68	72	0.944
## 36 Control	HO_MT	6	68	72	0.944
## 37 Mechanical disruption	HV_GAL	1	30	72	0.417
## 38 Mechanical disruption	HV_GAL	2	40	72	0.556
## 39 Mechanical disruption	HV_GAL	3	54	72	0.75
## 40 Mechanical disruption	HV_GAL	4	54	72	0.75
## 41 Mechanical disruption	HV_GAL	5	62	72	0.861
## 42 Mechanical disruption	HV_GAL	6	63	72	0.875
## 43 Mechanical disruption	HC_MT	1	3	48	0.0625
## 44 Mechanical disruption	HC_MT	2	11	48	0.229
## 45 Mechanical disruption	HC_MT	3	37	48	0.771
## 46 Mechanical disruption	HC_MT	4	41	48	0.854
## 47 Mechanical disruption	HC_MT	5	43	48	0.896
## 48 Mechanical disruption	HC_MT	6	43	48	0.896
## 49 Mechanical disruption	HO_VLN	1	45	96	0.469
## 50 Mechanical disruption	HO_VLN	2	49	96	0.510
## 51 Mechanical disruption	HO_VLN	3	57	96	0.594
## 52 Mechanical disruption	HO_VLN	4	61	96	0.635
## 53 Mechanical disruption	HO_VLN	5	73	96	0.760
## 54 Mechanical disruption	HO_VLN	6	73	96	0.760
## 55 Mechanical disruption	HO_SPT	1	17	72	0.236
## 56 Mechanical disruption	HO_SPT	2	59	72	0.819
## 57 Mechanical disruption	HO_SPT	3	64	72	0.889
## 58 Mechanical disruption	HO_SPT	4	66	72	0.917
## 59 Mechanical disruption	HO_SPT	5	66	72	0.917
## 60 Mechanical disruption	HO_SPT	6	66	72	0.917
## 61 Mechanical disruption	HO_SPC	1	40	72	0.556
## 62 Mechanical disruption	HO_SPC	2	63	72	0.875
## 63 Mechanical disruption	HO_SPC	3	63	72	0.875
## 64 Mechanical disruption	HO_SPC	4	67	72	0.931

```
## 65 Mechanical disruption HO_SPC      5    70    72    0.972
## 66 Mechanical disruption HO_SPC      6    70    72    0.972
## 67 Mechanical disruption HO_MT       1    47    72    0.653
## 68 Mechanical disruption HO_MT       2    59    72    0.819
## 69 Mechanical disruption HO_MT       3    66    72    0.917
## 70 Mechanical disruption HO_MT       4    66    72    0.917
## 71 Mechanical disruption HO_MT       5    66    72    0.917
## 72 Mechanical disruption HO_MT       6    66    72    0.917
```

9.4 By week and tumoral state

Same method as above, but stratified by tumoral state.

```
FB_proportions_tum <- FB_df %>%
  dplyr::mutate(week_FB_event = ceiling(age_followup)) %>%
  dplyr::group_by(tum_state_early) %>%
  dplyr::summarise(total_N = n(),
    n1 = sum(week_FB_event <= 1 & FB_event == 1),
    n2 = sum(week_FB_event <= 2 & FB_event == 1),
    n3 = sum(week_FB_event <= 3 & FB_event == 1),
    n4 = sum(week_FB_event <= 4 & FB_event == 1),
    n5 = sum(week_FB_event <= 5 & FB_event == 1),
    n6 = sum(week_FB_event <= 6 & FB_event == 1),
    p1 = n1 / total_N,
    p2 = n2 / total_N,
    p3 = n3 / total_N,
    p4 = n4 / total_N,
    p5 = n5 / total_N,
    p6 = n6 / total_N,
    .groups = "drop") %>%
  tidyr::pivot_longer(cols = matches("[np][1-6]$"),
    names_to = c(".value", "week"),
    names_pattern = "(n|p)([1-6])") %>%
  dplyr::mutate(week = as.numeric(week)) %>%
  dplyr::select(tum_state_early, week, n, total_N, raw_proportion = p)

print(FB_proportions_tum)
```

```
## # A tibble: 12 x 5
##   tum_state_early week      n total_N raw_proportion
##   <fct>          <dbl> <int>   <int>         <dbl>
## 1 Healthy        1    368    823         0.447
## 2 Healthy        2    586    823         0.712
## 3 Healthy        3    706    823         0.858
## 4 Healthy        4    724    823         0.880
## 5 Healthy        5    756    823         0.919
## 6 Healthy        6    764    823         0.928
## 7 Tumoral        1      1     41         0.0244
## 8 Tumoral        2     13     41         0.317
## 9 Tumoral        3     21     41         0.512
## 10 Tumoral       4     23     41         0.561
## 11 Tumoral       5     26     41         0.634
## 12 Tumoral       6     26     41         0.634
```

10 Plots of cumulative incidence of first bud production

10.1 Common themes

Theme of y-axis, and its presence or absence on the plots

```
scale_y <- scale_y_continuous(limits = c(0, 1), breaks = seq(0, 1, by = 0.25),
                              labels = percent, position = "right")
```

Fix to hide y-axis numbers, ticks, and titles on the right and left

```
hide_y <- theme(axis.text.y.right = element_blank(),
                axis.ticks.y.right = element_blank(),
                axis.title.y.right = element_blank(),
                axis.text.y = element_blank(),
                axis.ticks.y = element_blank(),
                axis.title.y = element_blank())
```

Fix to show y-axis numbers and ticks on the right

```
show_y <- theme(axis.text.y.right = element_text(color = "black", size = 14),
                axis.ticks.y.right = element_line(),
                axis.text.y = element_blank(),
                axis.ticks.y = element_blank(),
                axis.title.y = element_blank())
```

Theme of x-axis, and its presence or absence on the plots.

```
scale_x <- scale_x_continuous(limits = c(0, 6), breaks = seq(0, 6, by = 2))

hide_x <- theme(axis.text.x = element_blank(),
                axis.ticks.x = element_blank(),
                axis.title.x = element_blank())

show_x <- theme(axis.text.x = element_text(color = "black", size = 14),
                axis.ticks.x = element_line())
```

Color scale for tumoral state

```
tum_scale <- list(
  scale_color_manual(values = tum_colors),
  scale_fill_manual(values = tum_colors))
```

Color scales for rest condition

```
rest_scale <- list(
  scale_color_manual(values = rest_condition_colors, breaks = c("Control", "Mechanical
disruption")),
  scale_fill_manual(values = rest_condition_colors, breaks = c("Control", "Mechanical
disruption")))
```

Theme shared by panels A, B, and C (angle = 90 for bottom-to-top reading)

```
theme_common <- theme(  
  axis.title.x = element_text(color = "black", size = 14, margin = margin(t = 10)),  
  axis.title.y.right = element_text(color = "black", size = 14, angle = 90, margin =  
    margin(l = 15)),  
  plot.margin = margin(l = 5, r = 5, t = 5, b = 5))
```

Specific label configurations for precise placement

```
labs_base <- labs(x = NULL, y = NULL, title = NULL)  
labs_x_only <- labs(x = "Time (weeks)", y = NULL, title = NULL)  
labs_y_only <- labs(x = NULL, y = "Cumulative incidence of\nfirst bud production", title  
  = NULL)  
labs_xy <- labs(x = "Time (weeks)", y = "Cumulative incidence of\nfirst bud  
  production", title = NULL)
```

10.2 Panel A: interaction between rest condition and lineage

10.2.1 HO_SPT

```
A1 <- cuminc(Surv(age_followup, factor(status_competing)) ~ rest_condition,  
  data = FB_df %>% filter(lineage == "HO_SPT") %>%  
    mutate(rest_condition = factor(rest_condition, levels = c("Control",  
      "Mechanical disruption")))) %>%  
  ggcuminc(outcome = "1") +  
  add_confidence_interval() +  
  labs_base + scale_y + scale_x + rest_scale +  
  theme_bw() + theme_common + hide_y +  
  theme(axis.text.x = element_text(color = "black", size = 14),  
    legend.position = "none") +  
  annotate("text", x = 5.8, y = 0.05, label = expression(bolditalic("Ho") *  
    bold("_SPT")),  
    color = "black", size = 5, hjust = 1, vjust = 0)
```

10.2.2 HO_VLN

```
A2 <- cuminc(Surv(age_followup, factor(status_competing)) ~ rest_condition,  
  data = FB_df %>% filter(lineage == "HO_VLN") %>%  
    mutate(rest_condition = factor(rest_condition, levels = c("Control",  
      "Mechanical disruption")))) %>%  
  ggcuminc(outcome = "1") +  
  add_confidence_interval() +  
  labs_x_only + scale_y + scale_x + rest_scale +  
  theme_bw() + theme_common + hide_y +  
  theme(axis.text.x = element_text(color = "black", size = 14),  
    legend.position = "none") +  
  annotate("text", x = 5.8, y = 0.05, label = expression(bolditalic("Ho") *  
    bold("_VLN")),  
    color = "black", size = 5, hjust = 1, vjust = 0)
```

10.2.3 HC_MT

```
A3 <- cuminc(Surv(age_followup, factor(status_competing)) ~ rest_condition,
             data = FB_df %>% filter(lineage == "HC_MT") %>%
             mutate(rest_condition = factor(rest_condition, levels = c("Control",
                               "Mechanical disruption")))) %>%
  ggcuminc(outcome = "1") +
  add_confidence_interval() +
  labs_y_only + scale_y + scale_x + rest_scale +
  theme_bw() + theme_common + show_y +
  theme(axis.text.x = element_text(color = "black", size = 14),
        legend.position = "none") +
  annotate("text", x = 5.8, y = 0.05, label = expression(bolditalic("Hc") * bold("_MT")),
         color = "black", size = 5, hjust = 1, vjust = 0)
```

Extract legend from the plot A3

```
shared_treatment_legend <- cowplot::get_legend(
  A3 + theme(legend.position = "bottom",
            legend.direction = "horizontal",
            legend.text = element_text(size = 13),
            legend.title = element_text(size = 14, face = "bold")))
```

Combine A1, A2, and A3 into panel “A”

```
A <- (A1 + A2 + A3 + plot_layout(nrow = 1)) +
  plot_annotation(title = "Interaction between mechanical disruption and lineage",
                 theme = theme(plot.title = element_text(hjust = 0.5, face = "bold",
                 size = 16, margin = margin(b = 10))))
```

10.3 Panel B: by tumoral state

Cumulative incidence of first bud production by tumoral state.

```
B <- cuminc(Surv(age_followup, factor(status_competing)) ~ tum_state_early, data = FB_df)
%>%
  ggcuminc(outcome = "1") +
  add_confidence_interval() +
  labs_xy + scale_y + tum_scale + scale_x +
  theme_bw() + theme_common + show_y +
  theme(axis.text.x = element_text(color = "black", size = 14),
        axis.text.y.right = element_text(color = "black", size = 14),
        axis.ticks.y.right = element_line(),
        legend.position = "none") +
  plot_annotation(title = "Tumoral state effect",
                 theme = theme(plot.title = element_text(hjust = 0.5, face = "bold",
                 size = 16, margin = margin(b = 10))))
```

Extract legend from this panel

```
shared_tumor_legend <- cowplot::get_legend(
  B + theme(legend.position = "bottom",
    legend.direction = "horizontal",
    legend.text = element_text(size = 13),
    legend.title = element_text(size = 14, face = "bold")))
```

10.4 Panel C: by lineage

10.4.1 HO_MT

```
C1 <- cuminc(Surv(age_followup, factor(status_competing)) ~ 1,
  data = FB_df %>% filter(lineage == "HO_MT")) %>%
ggcuminc(outcome = "1") +
add_confidence_interval() +
labs_base + scale_y + scale_x +
theme_bw() + theme_common + hide_y + hide_x +
annotate("text", x = 5.8, y = 0.05, label = expression(bolditalic("Ho") * bold("_MT")),
  color = "black", size = 5, hjust = 1, vjust = 0)
```

10.4.2 HO_SPC

```
C2 <- cuminc(Surv(age_followup, factor(status_competing)) ~ 1,
  data = FB_df %>% filter(lineage == "HO_SPC")) %>%
ggcuminc(outcome = "1") +
add_confidence_interval() +
labs_base + scale_y + scale_x +
theme_bw() + theme_common + hide_y + hide_x +
annotate("text", x = 5.8, y = 0.05, label = expression(bolditalic("Ho") *
  bold("_SPC")),
  color = "black", size = 5, hjust = 1, vjust = 0)
```

10.4.3 HO_SPT

```
C3 <- cuminc(Surv(age_followup, factor(status_competing)) ~ 1,
  data = FB_df %>% filter(lineage == "HO_SPT")) %>%
ggcuminc(outcome = "1") +
add_confidence_interval() +
labs_base + scale_y + scale_x +
theme_bw() + theme_common + hide_y + hide_x +
theme(axis.text.y.right = element_text(color = "black", size = 14),
  axis.ticks.y.right = element_line()) +
annotate("text", x = 5.8, y = 0.05, label = expression(bolditalic("Ho") *
  bold("_SPT")),
  color = "black", size = 5, hjust = 1, vjust = 0)
```

10.4.4 HO_VLN

```
C4 <- cuminc(Surv(age_followup, factor(status_competing)) ~ 1,
             data = FB_df %>% filter(lineage == "HO_VLN")) %>%
  ggcuminc(outcome = "1") +
  add_confidence_interval() +
  labs_base + scale_y + scale_x +
  theme_bw() + theme_common + hide_y + show_x +
  annotate("text", x = 5.8, y = 0.05, label = expression(bolditalic("Ho") *
    bold("_VLN")),
           color = "black", size = 5, hjust = 1, vjust = 0)
```

10.4.5 HC_MT

```
C5 <- cuminc(Surv(age_followup, factor(status_competing)) ~ 1,
             data = FB_df %>% filter(lineage == "HC_MT")) %>%
  ggcuminc(outcome = "1") +
  add_confidence_interval() +
  labs_x_only + scale_y + scale_x +
  theme_bw() + theme_common + hide_y + show_x +
  annotate("text", x = 5.8, y = 0.05, label = expression(bolditalic("Hc") * bold("_MT")),
           color = "black", size = 5, hjust = 1, vjust = 0)
```

10.4.6 HV_GAL

```
C6 <- cuminc(Surv(age_followup, factor(status_competing)) ~ 1,
             data = FB_df %>% filter(lineage == "HV_GAL")) %>%
  ggcuminc(outcome = "1") +
  add_confidence_interval() +
  labs_y_only + scale_y + scale_x +
  theme_bw() + theme_common + show_y + show_x +
  annotate("text", x = 5.8, y = 0.05, label = expression(bolditalic("Hv") *
    bold("_GAL")),
           color = "black", size = 5, hjust = 1, vjust = 0)
```

Create panel C combining plots C1 to C6.

```
C <- (C1 + C2 + C3) / (C4 + C5 + C6) +
  plot_layout(heights = c(1, 1)) +
  plot_annotation(title = "Lineage effect",
                  theme = theme(plot.title = element_text(
                    hjust = 0.5, face = "bold", size = 16, margin = margin(b = 10))))
```

10.5 Assembly

Add a letter tag to the first sub-plot of each panel.

```
A[[1]] <- A[[1]] + labs(tag = "A") + theme(plot.tag = element_text(size = 20, face =
"bold"))
B <- B + labs(tag = "B") + theme(plot.tag = element_text(size = 20, face = "bold"))
C[[1]][[1]] <- C[[1]][[1]] + labs(tag = "C") + theme(plot.tag = element_text(size = 20,
face = "bold"))
```

Convert each panel into a graphical object.

```
A <- patchwork::patchworkGrob(A)
```

Wrap B in patchwork before conversion to force perfect axis alignment with A and C

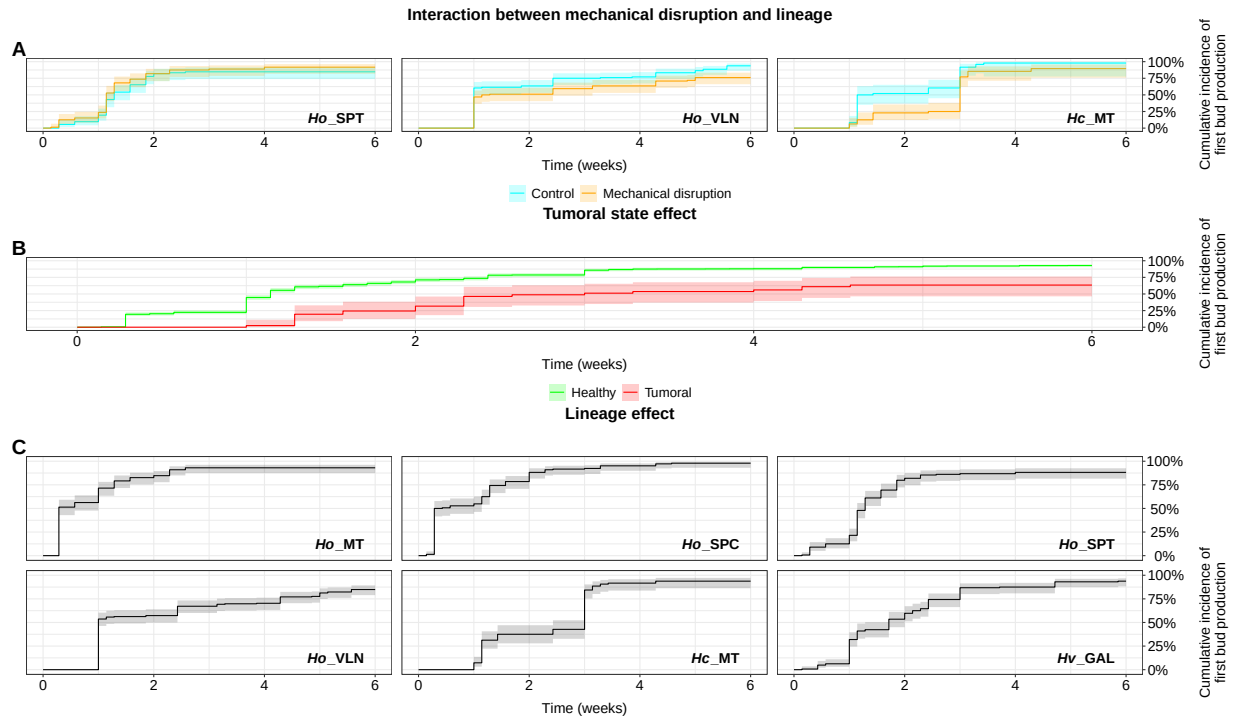
```
B <- patchwork::patchworkGrob(B + patchwork::plot_layout(ncol = 1))
C <- patchwork::patchworkGrob(C)
```

Combine panels and shared legends vertically into a single column, aligning axes.

```
final_FB_panel <- cowplot::plot_grid(A,
                                     shared_treatment_legend,
                                     B,
                                     shared_tumor_legend,
                                     C,
                                     ncol = 1,
                                     align = "v",
                                     axis = "lr",
                                     rel_heights = c(1, 0.1, 1, 0.1, 1.8)) +
  theme(plot.margin = margin(l = 5, r = 5, t = 5, b = 5))
```

Show the final figure.

```
print(final_FB_panel)
```



```
ggsave("figure_4.png", plot = final_FB_panel, width = 21, height = 29.7, units = "cm",
dpi = 300)
```